

B.Tech. Degree V Semester Examination, November 2009

ME 505 HEAT AND MASS TRANSFER

(2002 Scheme)

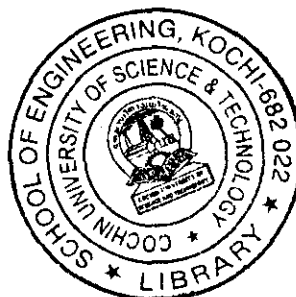
Time: 3 Hours

Maximum Marks: 100

- I. (a) What is meant by critical radius of insulation. Explain its significance. Derive an expression for the same in terms of thermal conductivity of insulating material and convective heat transfer coefficient of the surroundings. (10)
- (b) An exterior wall of a house may be approximated by a 5cm layer of common brick of thermal conductivity $0.7 \text{ W/m } ^\circ\text{C}$, followed by a 2 cm layer of gypsum plaster of thermal conductivity of $0.48 \text{ W/m } ^\circ\text{C}$. What thickness of loosely packed rock-wool insulation of thermal conductivity of $0.065 \text{ W/m } ^\circ\text{C}$ should be added to reduce the heat loss (or gain) through the wall by 80%? (10)
- OR**
- II. (a) Differentiate between fin effectiveness and fin efficiency. (5)
- (b) Derive an expression for the temperature distribution with respect to time, for lumped heat capacity method. Also mention the significance of non-dimensional numbers involved. (7)
- (c) A steel ball $[C_p = 0.46 \text{ kJ/kg } ^\circ\text{C}, \text{ and } k = 35 \text{ W/m } ^\circ\text{C}]$ of 5.0 cm diameter and initially at a uniform temperature of 450°C is suddenly placed in a controlled environment in which the temperature is maintained at 100°C . The convection heat transfer coefficient is $10 \text{ W/m}^2 \text{ } ^\circ\text{C}$. Calculate the time required for the ball to attain a temperature of 150°C . (8)
- III. (a) Explain hydrodynamic and thermal boundary layers with neat sketches. Also draw temperature profile and velocity profile over a flat plate. (10)
- (b) Air at 1 atm and 300K flows across a 20 cm^2 plate at a free-stream velocity of 20 m/s. The last half of the plate is heated to a constant temperature of 350K. Calculate the heat lost by the plate. (10)
- OR**
- IV. (a) Explain film boiling and pool boiling. What is meant by critical heat flux? What is its significance. (10)
- (b) Air at a pressure of 8 kN/m^2 and a temperature of 250°C flows over a flat plate of 0.3m width and 1m length at a velocity of 8 m/s. If the plate is to be maintained at a temperature of 78°C , estimate the rate of heat to be removed continuously from the plate. Using Reynold's-Colburn analogy estimate the drag force exerted on the plate if fluid flows over both surface of plate. (10)
- V. (a) Explain Kirchoff's law and from this law establish Kirchoff's identity. (10)
- (b) It is observed that the intensity of the radiation emitted by the sun is maximum at a wavelength of 0.5μ , Assuming sun to be a black body, estimate its surface temperature and emissive power. (10)

OR

(Turn Over)



- VI. (a) State and prove principle of conservation in case of radiation shape factor. (5)
 (b) Two parallel plates of size $1 \times 1 \text{ m}$ spaced 0.5 m apart are located in a very large room, the walls of which are maintained at a temperature of 27°C . One plate is maintained at a temperature of 900°C and the other at 400°C . Their emissivities are 0.2 and 0.5 respectively. If the plates exchange heat between themselves and surroundings, find the net heat transfer to each plate and to the room. Consider only the plate surfaces facing each other. (15)
- VII. (a) Classify heat exchangers depending upon
 (1) Direction of flow
 (2) Number of passes
 (3) Mixing of fluids
 Also show each one with a neat sketch. (10)
 (b) A counter flow concentric tube heat exchanger is used to cool engine oil of specific heat $2130 \text{ kJ/kg}^\circ \text{C}$ from 160°C to 60°C with water, available at 25°C as the cooling medium. The flow rate of cooling water through the inner tube of 0.5 m diameter is 2 kg/s , while the flow rate of oil through the outer annulus of outer diameter 0.7 m is also 2 kg/s . If the value of overall heat transfer coefficient is $250 \text{ W/m}^2\text{K}$, how long must be the heat exchanger to meet this requirement. (10)
- OR**
- VIII. (a) Explain condensers and evaporators as a special case heat exchangers. Also sketch the temperature distribution of each case. (8)
 (b) A 1-shell, 2-tube pass steam condenser consists of 3000 brass tubes of 20 mm diameter. Cooling water enters the tubes at 20°C with a mean flow rate of 3000 kg/s . The heat transfer coefficient for condensation on the outer surfaces of the tubes is $15,500 \text{ W/m}^2\text{K}$. If the heat load of the condenser is 2.3×10^8 when the steam condenses at 50°C . Determine
 (a) The outlet temperature of cooling water
 (b) The overall heat transfer coefficient
 (c) The tube length per pass using NTU method
 (d) The rate of condensation of steam if latent heat at that pressure is 2380 kJ/kg . (12)
- IX. (a) Explain Fick's law of diffusion with necessary terms and explanations. (10)
 (b) Discuss temperature and pressure dependence of diffusion coefficient. (10)
- OR**
- X. (a) Derive species conservation equation in diffusion mass transfer. (10)
 (b) O_2 gas at 25°C and a pressure of 2 bar is flowing through a rubber pipe of inside diameter 25 mm and wall thickness 2.5 mm . The diffusivity of O_2 through rubber is $0.21 \times 10^{-9} \text{ m}^2/\text{sec}$ and the solubility of O_2 in rubber is $3.12 \times 10^{-3} \text{ kmol/m}^3 \text{ bar}$. Find the loss of O_2 by diffusion per metre length of pipe? (10)
